Left-ideal cores upgrade asymptotic nonexpansiveness

Candidate substantial partial result for arXiv:2006.15393

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Status. Candidate substantial partial result, likely valid. The unrestricted Question 5.5 remains open here. We prove two transfer principles under which every one of the seven fixed-point statements named in that question does extend to asymptotically nonexpansive actions. One principle applies to noncompact amenable semigroups not covered by the compact-semigroup case in the source.

1 Source and exact question

The source is Bui Ngoc Muoi and Ngai-Ching Wong, *Fixed point theorems of various nonexpansive actions of semitopological semigroups on weakly/weak* compact convex sets*, arXiv:2006.15393v4 [1]. It proves fixed-point results for *super asymptotically nonexpansive* or *pointwise eventually nonexpansive* actions and closes with the question reproduced in Figure 1.

We end this paper with an open problem about possible extensions of our results. Under some conditions, we establish that a super asymptotically nonexpansive or pointwise eventually nonexpansive action of a semitopological semigroup S on weakly/weak* compact convex sets has a common fixed point.

Question 5.5. Do we have similar results as Theorems 3.1, 3.6, 4.1, 4.2, 4.4, and Propositions 3.11, 3.12 for asymptotically nonexpansive actions?

From Proposition 2.7, if S is compact and right reversible then the asymptotic nonexpansiveness coincides with the super asymptotic nonexpansiveness. Hence the question has an affirmative answer in this case.

Figure 1: Question 5.5 and the compact-semigroup case, from page 19 of the official arXiv PDF.

Transcription. "Do we have similar results as Theorems 3.1, 3.6, 4.1, 4.2, 4.4, and Propositions 3.11, 3.12 for asymptotically nonexpansive actions?" The source observes that the answer is affirmative when S is compact and right reversible.

2 Definitions

Let S be a semigroup acting on a subset K of a Banach space. A *left ideal* is a nonempty $I \subseteq S$ with $SI \subseteq I$; it is *supported by* $t \in S$ when $I \subseteq St$. The action is *asymptotically nonexpansive* if, for every $x, y \in K$, some left ideal $I_{x,y}$ satisfies

$$\|sx - sy\| \leq \|x - y\| \qquad (s \in I_{x,y}). \tag{2.1}$$

It is *super asymptotically nonexpansive* if, for every $x \in K$ and $t \in S$, some left ideal $J_{x,t} \subseteq St$ satisfies (2.1) for *every* $y \in K$. Omitting the support requirement gives pointwise eventual nonexpansiveness. For a locally convex space (E, Q), the Q-versions replace the norm by a specified $q \in Q$ and quantify over q.

A *least left ideal* is a left ideal L contained in every left ideal of S. For an infinite cardinal κ, say that S has property $(\mathrm{CI})_\kappa$ if every family of at most κ closed left ideals has nonempty intersection.

3 Proof intuition

The gap between the two action conditions is a quantifier gap. The weak condition gives one ideal for each pair (x, y); the strong condition needs a single ideal that works for all y and also lies below any prescribed principal ideal St.

There are two direct ways to close that gap. If S has a least left ideal, the same algebraic core lies in every $I_{x,y}$ and every St, so no topology or limiting argument is needed. More generally, intersect the closed ideals belonging to a dense set of y's together with St. Property $(\mathrm{CI})_\kappa$ makes this intersection nonempty, and separate weak continuity plus weak lower semicontinuity extends the inequalities from the dense set to all of K.

Even without either hypothesis, right reversibility lets one satisfy any *finite* collection of pair inequalities. Compactness of K^K then produces a globally nonexpansive pointwise limit map in the closure of every principal tail. The missing preservation of weak continuity and surjectivity pinpoints why this last observation does not settle the full question.

4 The least-left-ideal transfer principle

Theorem 1 (algebraic upgrade)**.** *If S has a least left ideal L, every asymptotically nonexpansive action of S is super asymptotically nonexpansive. The same assertion holds for asymptotically Q-nonexpansive actions on locally convex spaces. No topology or continuity on S is required.*

Proof. Fix $x \in K$ and $t \in S$. For every $y \in K$, asymptotic nonexpansiveness supplies a left ideal $I_{x,y}$. Minimality of L gives $L \subseteq I_{x,y}$ for every y. Moreover, St is a nonempty left ideal, so $L \subseteq St$. Thus L itself is supported by t and every $s \in L$ satisfies (2.1) simultaneously for all y. Taking $J_{x,t} = L$ proves the claim. The proof for a seminorm q is identical, separately for each q. □

Corollary 2 (all seven source results)**.** *In each of Theorems 3.1, 3.6, 4.1, 4.2, 4.4 and Propositions 3.11, 3.12 of [1], suppose in addition that S has a least left ideal. Then the source's super asymptotic or pointwise eventual hypothesis may be replaced by asymptotic nonexpansiveness (and by asymptotic Q-nonexpansiveness in Theorem 4.4).*

Proof. Theorem 1 supplies the original action hypothesis in every case. Apply the corresponding theorem or proposition of [1] without altering any of its remaining assumptions. □

Proposition 3 (a noncompact amenable family)**.** *Let G be any infinite discrete amenable group and put*

$$S = G \times \{0, 1\}, \qquad (g, i)(h, j) = (gh, \min\{i, j\}).$$

Then S is a noncompact discrete amenable semitopological semigroup and $L = G \times \{0\}$ is its least left ideal. In particular, Theorem 3.1 of [1] holds for asymptotically nonexpansive actions of this S.

Proof. If a left ideal I contains (h, j), then left multiplication by all $(g, 0)$ shows that $G \times \{0\} \subseteq I$. The latter set is itself a left ideal, hence it is least. The product is discrete and noncompact. The projection $S \to G$ and the two-point semilattice factor show amenability; equivalently, take the product of invariant means. All left ideals contain L, so S is right reversible. Corollary 2 now applies. □

For the completely explicit instance $G = \mathbb{Z}$, the semigroup is also commutative; its principal left ideals are L on layer 0 and S on layer 1. The verifier enumerates the finite quotients $\mathbb{Z}/n\mathbb{Z} \times \{0, 1\}$ and checks the least-ideal statement exhaustively.

5 A density/intersection transfer principle

Theorem 4 (closed-ideal upgrade)**.** *Let S be a semitopological semigroup, let K be a subset of a locally convex space E, and let $D \subseteq K$ be dense in the original locally convex topology with $|D| \leq \kappa$, where κ is infinite. Assume:*

(i) every principal left ideal St is closed;

(ii) S has $(\mathrm{CI})_\kappa$*;*

(iii) the action is separately weakly continuous.

Then every asymptotically Q-nonexpansive action is super asymptotically Q-nonexpansive. In a Banach space it is enough that D be norm dense, and the conclusion applies to ordinary asymptotic nonexpansiveness.

Proof. Fix $x \in K$, $t \in S$, and a continuous seminorm q. For each $y \in D$ choose a left ideal I_y on which

$$q(sx - sy) \leq q(x - y). \tag{5.1}$$

Its closure $\overline{I}_y$ is a closed left ideal: for $a \in S$, separate continuity of multiplication gives $a\overline{I}_y \subseteq \overline{(aI_y)} \subseteq \overline{I}_y$. Moreover (5.1) remains true for every $s \in \overline{I}_y$. Indeed, along a net $s_\alpha \to s$, separate weak continuity gives $s_\alpha x \to sx$ and $s_\alpha y \to sy$ weakly, while every continuous seminorm is weakly lower semicontinuous.

The family

$$\{St\} \cup \{\overline{I}_y : y \in D\}$$

contains at most κ closed left ideals. Hence its intersection J is nonempty by $(\mathrm{CI})_\kappa$. It is a left ideal, lies in St, and every $s \in J$ satisfies (5.1) for all $y \in D$.

For arbitrary $y \in K$, choose a net (y_α) in D converging to y in the original topology. Then $sy_\alpha \to sy$ weakly, and weak lower semicontinuity gives

$$q(sx - sy) \leq \liminf_\alpha q(sx - sy_\alpha) \leq \lim_\alpha q(x - y_\alpha) = q(x - y).$$

Thus J is the ideal required by super asymptotic Q-nonexpansiveness. $\square$

Remark 5. The norm-separable case of Proposition 2.7(b) in [1] is recovered by using right reversibility for the finite-intersection property, countable compactness to intersect the countable family, and C-closedness to ensure that St is closed. Theorem 4 isolates exactly the ideal-theoretic content needed at an arbitrary density and does not require compactness of S itself.

6 What right reversibility alone still gives

Write $T_s x = sx$ and give K^K the product of the weak topologies on a weakly compact K. Thus K^K is compact.

Lemma 6 (nonexpansive maps in every tail closure)**.** *Let S be right reversible and semitopological, and let its asymptotically nonexpansive action on a weakly compact K be separately weakly continuous. For every $t \in S$, the pointwise-weak closure*

$$\overline{\{T_s : s \in St\}}^{K^K}$$

contains a globally nonexpansive map $p_t : K \to K$.

Proof. Fix a finite $F \subseteq K \times K$. For each $(x, y) \in F$, close the ideal $I_{x,y}$. The sets $\overline{St}$ and $\overline{I}_{x,y}$ are closed left ideals. Right reversibility and induction give a point

$$r_F \in \overline{St} \cap \bigcap_{(x,y)\in F} \overline{I}_{x,y}.$$

Separate weak continuity and weak lower semicontinuity imply $\|r_F x - r_F y\| \leq \|x - y\|$ for every $(x, y) \in F$. Also T_{r_F} belongs to the pointwise-weak closure of $\{T_s : s \in St\}$, because $r_F \in \overline{St}$.

Direct the finite subsets F by inclusion. Compactness of K^K supplies a cluster point p_t of (T_{r_F}). For each fixed pair (x, y) the relevant inequality holds eventually, and weak lower semicontinuity passes it to the limit. Hence $\|p_t x - p_t y\| \leq \|x - y\|$ for all x, y. The tail closure is closed, so it contains p_t. □

7 Verification, limitations, and novelty

The proofs above use only four independently checkable facts: closures of left ideals remain left ideals in a semitopological semigroup; continuous seminorms are weakly lower semicontinuous; K^K is compact by Tychonoff; and right reversibility gives finite intersections of closed left ideals. The included verifier exhaustively checks Proposition 3 for finite cyclic quotients and reports the principal left ideals and every left ideal.

Eight focused upgrade attempts were made. Counterexample routes based on commuting interval maps and left-zero elements did not meet the simultaneous amenability, reversibility, continuity, and asymptotic inequalities. Direct adaptations of the source's minimal-set proof failed at the same uniformity step. Lemma 6 is the strongest unrestricted reduction found, but a map in a pointwise-weak closure need not be weakly continuous or surjective. Those failures prevent the preimage and minimality arguments in the source from being transferred. Accordingly, this packet makes no claim about Question 5.5 without the additional ideal hypotheses.

Bounded searches of the run indexes, the exact question text, and later work on strong/asymptotically nonexpansive actions found no explicit statement of Theorems 1–4 or a full resolution of Question 5.5. The arguments are elementary enough that they may be folklore; novelty confidence is therefore moderate. Mathematical confidence is high for the two transfer theorems and the compactification lemma. Human review should focus on the intended convention for right reversibility and on whether the least-left-ideal observation has appeared elsewhere under different terminology.

References

[1] B. N. Muoi and N.-C. Wong, *Fixed point theorems of various nonexpansive actions of semitopological semigroups on weakly/weak* compact convex sets*, arXiv:2006.15393v4 (2022).